Common Equations Used in Chemistry

John Terhorst, Ph.D.

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1. Basic Conversions and Definitions

Converting °F to °C:

$$°\mathrm{C} = (°\mathrm{F} - 32) \times \frac{5}{9}$$

Converting °C to °F:

$$°\mathrm{F} = \left(°\mathrm{C} \times \frac{9}{5}\right) + 32$$

Converting °C to K:

$$K = °\mathrm{C} + 273.15$$

2. Solutions

Density:

$$d = \frac{m}{V}$$

Molarity:

$$C = \frac{\text{moles of solute}}{\text{liters of solution}}$$

Molality:

$$b = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

Boiling point elevation[1]:

$$\Delta T_\mathrm{b} = iK_\mathrm{b}b$$

Freezing point depression:

$$\Delta T_\mathrm{f} = iK_\mathrm{f}b$$

Table 1: Boiling point elevation and freezing point depression constants for different solvents.

Solvent	Normal Boiling (°C)	K_b (°C/m)	Normal Freezing (°C)	K_f (°C/m)
Water	100.0	0.512	0.0	1.86
Acetic acid	118.1	3.04	16.6	3.90
Benzene	80.1	2.53	5.5	5.12
Chloroform	61.3	3.63	−63.5	4.86
Carbon tetrachloride	76.8	5.26	−22.62	31.4
Carbon disulfide	46.2	2.42	−112.1	3.74
Nitrobenzene	210.9	5.24	5.67	6.87
D-Camphor	207.4	4.91	178.8	37.8

Osmotic pressure:

$$\Pi = i\mathcal{M}RT$$

[1] The van't Hoff factor, i, describes the number of particles formed upon solvation. For example, for solvation of sodium chloride, $\mathrm{NaCl}(s) \longrightarrow \mathrm{Na}^+(aq) + \mathrm{Cl}^-(aq)$, $i = 2$. For a nonelectrolyte (like glucose or benzene), $i = 1$. This factor can deviate from ideality for highly charged ions, where ion-pairing occurs (incomplete dissociation), as is seen with $\mathrm{CaCl_2}$, where $i = 2.6$ instead of 3.0 ($\mathrm{CaCl_2}(s) \longrightarrow \mathrm{Ca}^{2+}(aq) + 2\mathrm{Cl}^-(aq)$).

Dilution:

$$C_1V_1 = C_2V_2$$

Dilution (Shortcut):[2]

$$C_2 = C_1\left(\frac{V_{\text{used}}}{V_{\text{tot}}}\right)$$

3. Gases

Ideal Gas Law, $R = 0.08206\ \frac{\text{L}\cdot\text{atm}}{\text{K}\cdot\text{mol}}$

$$PV = nRT$$

Boyle's Law (n, T constant):

$$P_1V_1 = P_2V_2$$

Charles' Law (n, P constant):

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

Avogadro's Law (P, T constant):

$$\frac{V_1}{V_2} = \frac{n_1}{n_2}$$

Gay-Lussac's Law (P, T constant):

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

Combined Gas Law (n is constant):

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Relation of density to molar mass:

$$\mathcal{M} = \frac{dRT}{P}$$

Dalton's Law of Partial Pressures, where χ_i = mole fraction of species i and P_{T} is total pressure:

$$P_i = \chi_i P_{\text{T}}$$

Raoult's Law, where $P_i^\star$ is the vapor pressure of pure substance i:

$$P_{\text{Tot}} = \chi_{\text{A}}P_{\text{A}}^\star + \chi_{\text{B}}P_{\text{Bb}}^\star + \ldots$$

For ionic solutes we include the van't Hoff factor i:

$$P_i = i\chi_i P_i^\star$$

Root-mean-square (RMS) speed of a gas particle:

$$\mu = \sqrt{\frac{3RT}{\mathcal{M}}}$$

[2] Say we use 11.3 mL of 0.030 M KI that is diluted to 50 mL. Moles of KI is 0.030 mol/L × 0.0113 L = 0.000339 mol, and therefore [KI] is 0.000339 mol / 0.050 L = 0.00678 M. Using this shortcut, $C_2 = 0.030 \times (11.3/50) = 0.00678$ M.

Rates of Effusion by Molar Mass

$$\frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$$

van der Waals equation for pressure of a non-ideal gas, where a corrects for the attractive forces between gas particles, and b corrects for the volume occupied by gas particles:

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

$$P = \frac{nRT}{(V - nb)} - \frac{n^2 a}{V^2}$$

4. Thermodynamics

4.1. Heat Exchange

Heat exchange, where C_s is specific heat capacity:

$$q = mC_s\Delta T$$

Calorimetry in an imperfect calorimeter:

$$q_{\text{rxn}} = mC_s\Delta T + C_{\text{cal}}\Delta T$$

Heat exchange, where C_V is molar heat capacity at constant volume:

$$q = nC_V\Delta T$$

Heat exchange, where C_P is molar heat capacity at constant pressure:

$$q = nC_P\Delta T$$

4.2. Conditions

4.2.1. Definitions

- Isothermal: $\Delta T = 0$, $\Delta E = 0$, $q = -w$
- Adiabatic: $q = 0$, $\Delta E = w$
- Isobaric: $\Delta P = 0$ (constant pressure)

4.2.2. In General

- $q = nC\Delta T$
- $w = -P\Delta V$ where L·atm = 101.325 J
- $\Delta E = q + w$
- $\Delta H = \Delta E + P\Delta V + V\Delta P$
- $C_{\text{P}} = \frac{5}{2}R$ and $C_{\text{V}} = \frac{3}{2}R$

Table 2: Summary of thermodynamic quantities under different conditions.

Quantity	Constant P	Constant V	Adiabatic	Isothermal
Heat, q	$nC_\mathrm{P}\Delta T$	$nC_\mathrm{V}\Delta T$	0	$-w$
Work, w	$-P\Delta V$	0	$nC_V\Delta T$	$-nRT\ln\left(\frac{V_\mathrm{f}}{V_\mathrm{i}}\right) = -nRT\ln\left(\frac{P_\mathrm{i}}{P_\mathrm{f}}\right)$
Energy, ΔE	$q+w$	q	w	0
Enthalpy, ΔH	q	$\Delta E + V\Delta P$	$V\Delta P$	0
Entropy, ΔS	$nC_\mathrm{P}\ln\left(\frac{T_\mathrm{f}}{T_\mathrm{i}}\right)$	$nC_\mathrm{V}\ln\left(\frac{T_\mathrm{f}}{T_\mathrm{i}}\right)$	0	$nR\ln\left(\frac{V_\mathrm{f}}{V_\mathrm{i}}\right) = nR\ln\left(\frac{P_\mathrm{i}}{P_\mathrm{f}}\right)$

4.3. The Adiabatic Condition

Given a rapid change in volume where $q = 0$, both P and T change to accommodate the compression or expansion.

$$\gamma = \frac{C_P}{C_V} = \frac{5}{3}$$

$$(P_1)(V_1)^\gamma = (P_2)(V_2)^\gamma$$

$$w = nC_V\Delta T$$

Temperature change in adiabatic compression or expansion:

$$T_\mathrm{f} = T_\mathrm{i}\left(\frac{V_\mathrm{i}}{V_\mathrm{f}}\right)^{(R/C_V)}$$

4.4. Enthalpy

Enthalpy is the energy required to create a system, plus the amount of energy required to make room for the system by displacing its environment by the system's volume at a given pressure.

$$H = E + PV$$

Enthalpy change for a reaction:

$$\Delta H = \Delta E + \Delta(PV) = \Delta E + P\Delta V + V\Delta P$$

Standard enthalpy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta H^\circ_\mathrm{rxn} = \sum n\Delta H^\circ_\mathrm{f}(\text{products}) - \sum m\Delta H^\circ_\mathrm{f}(\text{reactants}) = \sum D_\mathrm{broken} - \sum D_\mathrm{formed}$$

Clausius–Clapeyron equation:

$$\ln P = \frac{-\Delta H_\mathrm{vap}}{RT} + C$$

Heat of vaporization based on the Clausius–Clapeyron equation:

$$\ln\frac{P_1}{P_2} = \frac{\Delta H_\mathrm{vap}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{-\Delta H_\mathrm{vap}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

4.5. Entropy

Second Law of Thermodynamics:

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$$

Standard entropy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta S^{\circ}_{\text{rxn}} = \sum n\Delta S^{\circ}(\text{products}) - \sum m\Delta S^{\circ}(\text{reactants})$$

Change in entropy with ΔH_{vap} or q where T is constant:

$$\Delta S = \frac{\Delta H_{\text{vap}}}{T} = \frac{q}{T}$$

Change in entropy with ΔV or ΔP where T is constant:

$$\Delta S = nR\ln\left(\frac{V_{\text{f}}}{V_{\text{i}}}\right) = nR\ln\left(\frac{P_{\text{i}}}{P_{\text{f}}}\right)$$

Change in entropy for a thermodynamic process where T and/or P are varied:

$$\Delta S = nC_V\ln\left(\frac{T_{\text{f}}}{T_{\text{i}}}\right) - nR\ln\left(\frac{P_{\text{f}}}{P_{\text{i}}}\right)$$

4.6. Gibbs Free Energy

Free energy change at constant temperature:

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

Or for an equilibrium reaction:

$$\Delta G^{\circ} = -RT\ln K_{\text{eq}}$$

Non-standard free energy:

$$\Delta G = \Delta G^{\circ} + RT\ln Q$$

Or:

$$\Delta G = RT\ln\left(\frac{Q}{K}\right)$$

Standard free energy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta G^{\circ}_{\text{rxn}} = \sum n\Delta G^{\circ}_{\text{f}}(\text{products}) - \sum m\Delta G^{\circ}_{\text{f}}(\text{reactants})$$

5. Electromagnetism

Relationship of wavelength and frequency, where $c = 2.99 \times 10^8$ m/s:

$$c = \lambda\nu$$

Energy of a photon, where Planck's constant $h = 6.626 \times 10^{-34}$ J·s:

$$E = h\nu = \frac{hc}{\lambda}$$

Energy of an electron in the nth shell in a hydrogen atom, where $R_\infty = 2.18 \times 10^{-18}$ J:

$$E_n = Z^2 R_\infty \left(\frac{1}{n^2}\right)$$

Energy of a photon corresponding to an electron transition from the n_i shell to the n_f shell:

$$\Delta E = -Z^2 R_\infty \left(\frac{1}{n_\text{f}^2} - \frac{1}{n_\text{i}^2}\right)$$

Or, using $R_\infty = 10973731.6\ \text{m}^{-1}$:

$$\frac{1}{\lambda} = -Z^2 R_\infty \left(\frac{1}{n_\text{f}^2} - \frac{1}{n_\text{i}^2}\right)$$

de Broglie wavelength; if the particle size is much smaller than λ, then quantum rules are needed:

$$\lambda = \frac{h}{mv}$$

Formal charge:

$$q_\text{F} = N_i e_\text{V} - N_j \text{B} - N_k e_\text{NB}$$

Dipole moment:

$$\mu = Q \times r$$

Bragg equation:

$$\lambda\nu = 2d \sin\theta$$

Coulomb's law, where k_e is Coulomb's constant $k_\text{e} \approx 8.99 \times 10^9\ \text{N}\,\text{m}^2\,\text{C}^{-2}$

$$F = k_\text{e} \frac{q_1 q_2}{r^2}$$

cf. Newton's law of gravitation, where G is the universal gravitational constant, $6.674 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$

$$F = G \frac{m_1 m_2}{r^2}$$

6. Quantum Mechanics

Heisenberg Uncertainty Principle, Δx is the minimum "size of a box" for a particle:

$$\Delta x \Delta \rho \geq \frac{h}{4\pi}$$

The time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

The time-independent Schrödinger equation for a particle of mass m moving in one direction with energy E where $\hbar = h/2\pi$ is:

$$-\frac{\hbar^2}{2m} \frac{d^2\Psi(x)}{dx^2} + V(x)\Psi(x) = E\Psi(x)$$

Wavefunction for the $1s$ orbital:

$$\psi_{1s} = \frac{1}{\sqrt{\pi}} e^{-r}$$

Wavefunction for the $2s$ orbital:

$$\psi_{2s} = \frac{1}{2\sqrt{2\pi}} \left(1 - \frac{r}{2}\right) e^{-r/2}$$

Wavefunction for the $2p_z$ orbital, where a_0 is the radius of the first Bohr orbit (5.29×10^{-11} m), $\sigma = Z(r/a_0)$ where r is the distance (in m) from the nucleus and Z is the nuclear charge, and θ is an angle:

$$\psi_{2p_z} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0}\right)^{3/2} \sigma e^{-\sigma/2} \cos\theta$$

Wavefunction for a particle in a 1-dimensional box of length L:

$$\psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}\right) x$$

Energy for a particle in a 1-dimensional box of length L:

$$E = \frac{n^2 h^2}{8mL^2}$$

Potential energy between two charged bodies:

$$V = k\frac{q_1 q_2}{r^2}$$

For a particle in three dimensions, using spherical polar coordinates:

$$\Psi(\rho, \theta, \phi) = R(r) \times \Theta(\theta) \times \Phi(\phi)$$

6.1. 3-Dimensional H-Atom Wavefunctions

$R_{n\ell}(r)$	$\Theta_{\ell m}(\theta)$	$\Phi_m(\phi)$
$\langle n, \ell\rangle = \langle 1, 0\rangle\,(1s)$	$\langle \ell, m\rangle = \langle 0, 0\rangle\,(s)$	$\langle m\rangle = \langle 0\rangle\,(s, z, z^2)$
$(Z/a_0)^{3/2} \cdot 2e^{-\rho/2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{\sqrt{2\pi}}$
$\langle n, \ell\rangle = \langle 2, 0\rangle\,(2s)$	$\langle \ell, m\rangle = \langle 1, 0\rangle\,(p_z)$	$\langle m\rangle = \langle 1\cos\rangle\,(x, xz)$
$\frac{(Z/a_0)^{3/2}}{2\sqrt{2}} \cdot (2-\rho)e^{-\rho/2}$	$\frac{\sqrt{6}}{2}\cos(\theta)$	$\frac{1}{\sqrt{\pi}}\cos(\phi)$
$\langle n, \ell\rangle = \langle 2, 1\rangle\,(2p)$	$\langle \ell, m\rangle = \langle 1, 1\rangle\,(p_{x,y})$	$\langle m\rangle = \langle 1\sin\rangle\,(y, yz)$
$\frac{(Z/a_0)^{3/2}}{2\sqrt{6}} \cdot \rho e^{-\rho/2}$	$\frac{\sqrt{3}}{2}\sin(\theta)$	$\frac{1}{\sqrt{\pi}}\sin(\phi)$
$\langle n, \ell\rangle = \langle 3, 0\rangle\,(3s)$	$\langle \ell, m\rangle = \langle 2, 0\rangle\,(d_{z^2})$	$\langle m\rangle = \langle 2\cos\rangle\,(x^2 - y^2)$
$\frac{(Z/a_0)^{3/2}}{9\sqrt{3}} \cdot (6 - 6\rho + \rho^2)e^{-\rho/2}$	$\frac{\sqrt{10}}{4}\left(3\cos^2(\theta) - 1\right)$	$\frac{1}{\sqrt{\pi}}\cos(2\phi)$
$\langle n, \ell\rangle = \langle 3, 1\rangle\,(3p)$	$\langle \ell, m\rangle = \langle 2, 1\rangle\,(d_{xz}, d_{yz})$	$\langle m\rangle = \langle 2\sin\rangle\,(xy)$
$\frac{(Z/a_0)^{3/2}}{9\sqrt{6}} \cdot (4-\rho)e^{-\rho/2}$	$\frac{\sqrt{15}}{2}\left(\sin(\theta)\cos(\theta)\right)$	$\frac{1}{\sqrt{\pi}}\sin(2\phi)$
$\langle n, \ell\rangle = \langle 3, 2\rangle\,(3d)$	$\langle \ell, m\rangle = \langle 2, 2\rangle\,(d_{xy}, d_{x^2-y^2})$	
$\frac{(Z/a_0)^{3/2}}{9\sqrt{30}} \cdot \rho^2 e^{-\rho/2}$	$\frac{\sqrt{15}}{2}\sin^2(\theta)$	

7. Stoichiometry

Percent composition of an element in a compound, where n = the number of moles of the element in one mole of the compound:

$$\% = \left(\frac{n \times \text{molar mass of element}}{\text{molar mass of compound}}\right) \times 100\%$$

Percent yield:

$$\%\ \text{yield} = \left(\frac{\text{actual yield}}{\text{theoretical yield}}\right) \times 100\%$$

8. Electrochemistry

Electrical force between two charged bodies:

$$F_{e\ell} = k\frac{q_1 q_2}{r_2}$$

Standard emf of an electrochemical cell $F = 96485$ coulomb/mol:

$$\mathscr{E}^\circ_{\text{cell}} = \mathscr{E}^\circ_{\text{ox}} - \mathscr{E}^\circ_{\text{red}} = \frac{RT}{nF}\ln K$$

Standard free energy of an electrochemical cell:

$$\Delta G^\circ = -nF\mathscr{E}^\circ_{\text{cell}}$$

Or:

$$\mathscr{E}_{\text{cell}} = \left(\frac{\Delta S}{nF}\right)T + \left(\frac{-\Delta H}{nF}\right)$$

Nernst equation:

$$\mathscr{E}_{\text{cell}} = \mathscr{E}^\circ_{\text{cell}} - \frac{RT}{nF}\ln Q$$

$$\mathscr{E}_{\text{cell}} = \mathscr{E}^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log Q$$

At 298 K:

$$\mathscr{E}_{\text{cell}} = \mathscr{E}^\circ_{\text{cell}} - \frac{0.02569}{n}\ln Q$$

$$\mathscr{E}_{\text{cell}} = \mathscr{E}^\circ_{\text{cell}} - \frac{0.0592}{n}\log Q$$

Standard free energy of an electrochemical cell:

$$\Delta G = -nF\mathscr{E}_{\text{cell}}$$

Reduction of metal with amps, where n is number of electrons per mol of reactant:

$$m = \frac{A \cdot t \cdot \mathcal{M}}{n \cdot F}$$

9. Reaction Kinetics

	Rate Laws		
Order	Standard Form	Integrated Form	Line Form
0	rate $= k$	$[\text{A}]_t - [\text{A}]_0 = -kt$	$[\text{A}]_t = -kt + [\text{A}]_0$
1	rate $= k[\text{A}]$	$\ln \frac{[\text{A}]_t}{[\text{A}]_0} = -kt$	$\ln [\text{A}]_t = -kt + \ln [\text{A}]_0$
2	rate $= k[\text{A}]^2$	$\frac{1}{[\text{A}]_t} - \frac{1}{[\text{A}]_0} = kt$	$\frac{1}{[\text{A}]_t} = kt + \frac{1}{[\text{A}]_0}$

Order	Half-life	Alternate Form
0	$t_{1/2} = \frac{[\text{A}]_0}{2k}$	
1	$t_{1/2} = \frac{\ln 2}{k}$	$\ln \frac{[\text{A}]_t}{[\text{A}]_0} = (-\ln 2)\left(\frac{t}{t_{1/2}}\right)$
2	$t_{1/2} = \frac{1}{k[\text{A}]_0}$	

Arrhenius equation:

$$k = A \exp \frac{-E_{\mathrm{a}}}{RT} \text{ and ln form: } \ln k = \left(\frac{-E_{\mathrm{a}}}{R}\right)\left(\frac{1}{T}\right) + \ln A$$

Arrhenius equation for different rate constants for different E_{a} at constant T:

$$\ln\left(\frac{k_2}{k_1}\right) = \left(\frac{1}{RT}\right)(-E_{\mathrm{a}_2} + E_{\mathrm{a}_1})$$

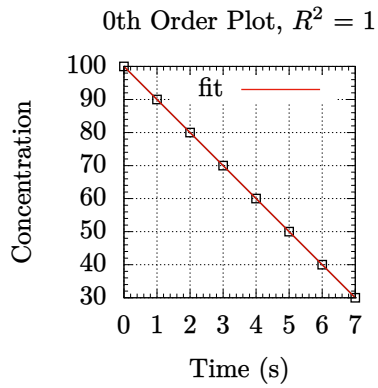
Relationships of rate constants at two different temperatures:

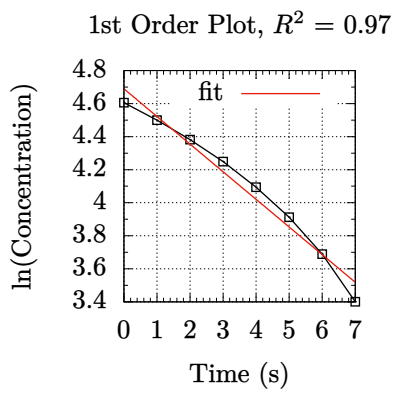
$$\ln \frac{k_1}{k_2} = \frac{E_{\mathrm{a}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{-E_{\mathrm{a}}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

9.1. Kinetics Data

Table 3: Kinetics Data — 0th Order

Time	Concentration	ln(Concentration)	1/(Concentration)
0	100	4.605	0.0100
1	90	4.499	0.0110
2	80	4.382	0.0125
3	70	4.248	0.0141
4	60	4.094	0.0166
5	50	3.912	0.0200
6	40	3.688	0.0250
7	30	3.401	0.0333





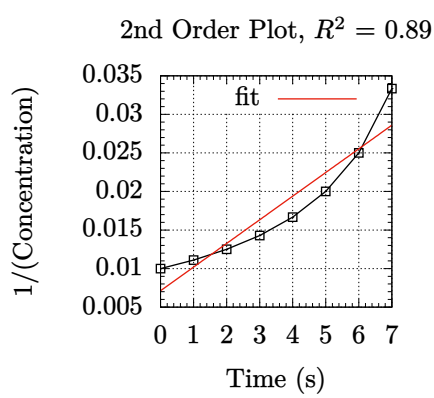
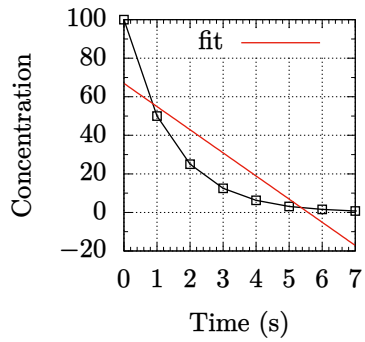


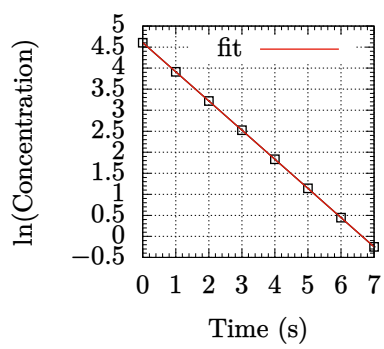
Table 4: Kinetics Data — 1st Order

Time	Concentration	ln(Concentration)	1/(Concentration)
0	100	4.605	0.01
1	50	3.912	0.02
2	25	3.218	0.04
3	12.5	2.525	0.08
4	6.25	1.832	0.16
5	3.13	1.141	0.32
6	1.56	0.444	0.64
7	0.78	−0.248	1.28

0th Order Plot, $R^2 = 0.73$



1st Order Plot, $R^2 = 1$



2nd Order Plot, $R^2 = 0.72$

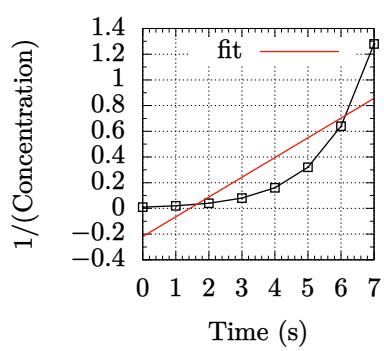
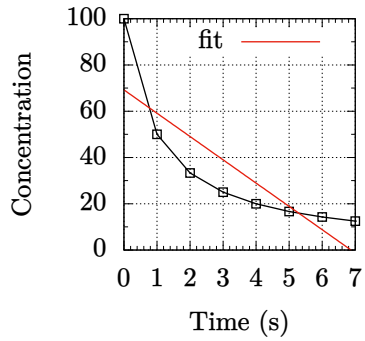


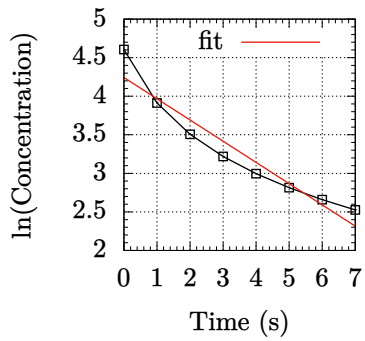
Table 5: Kinetics Data — 2nd Order

Time	Concentration	ln(Concentration)	1/(Concentration)
0	100	4.605	0.01
1	50	3.912	0.02
2	33.3	3.505	0.03
3	25	3.218	0.04
4	20	2.995	0.05
5	16.67	2.813	0.06
6	14.28	2.658	0.07
7	12.5	2.525	0.08

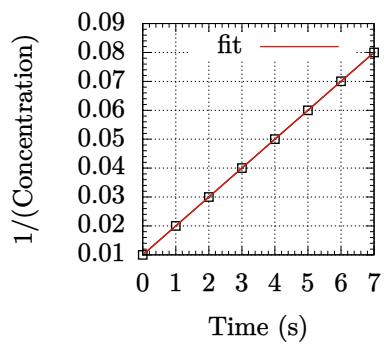
0th Order Plot, $R^2 = 0.70$



1st Order Plot, $R^2 = 0.92$



2nd Order Plot, $R^2 = 1$



10. Equilibrium

Law of Mass Action (equilibrium constant) for a reaction of form $a\mathrm{A} + b\mathrm{B} \longrightarrow c\mathrm{C} + d\mathrm{D}$:

$$K_{\mathrm{eq}} = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

van't Hoff Equation

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{-\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \text{ or } \ln K_{\mathrm{sp}} = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Relationship between equilibrium constants for aqueous systems and for gases:

$$K_{\mathrm{p}} = K_{\mathrm{c}} RT^{\Delta n}$$

Ion product of water:

$$K_{\mathrm{w}} = [\mathrm{H_3O^+}][^-\mathrm{OH}] = 1 \times 10^{-14}$$

The Quadratic Formula: for a quadratic equation $ax^2 + bx + c = 0$, the solutions x are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

10.1. Equilibrium of Acids and Bases

Definition of pH:

$$\mathrm{pH} = -\log[\mathrm{H_3O^+}]$$

Definition of pOH:

$$\mathrm{pOH} = -\log[^-\mathrm{OH}] = 14 - \mathrm{pH}$$

For the reaction $\mathrm{HA} + \mathrm{H_2O} \longrightarrow \mathrm{H_3O^+} + \mathrm{A^-}$:

$$K_{\mathrm{a}} = \frac{[\mathrm{H_3O^+}][\mathrm{A^-}]}{[\mathrm{HA}]}$$

For the reaction $\mathrm{B} + \mathrm{H_2O} \longrightarrow \mathrm{BH^+} + {}^-\mathrm{OH}$:

$$K_{\mathrm{b}} = \frac{[\mathrm{BH^+}][^-\mathrm{OH}]}{[\mathrm{B}]}$$

Relationship between K_{a}, K_{b}, and K_{w}:

$$K_{\mathrm{w}} = K_{\mathrm{a}} K_{\mathrm{b}} = 1 \times 10^{-14}$$

Definition of $\mathrm{p}K_{\mathrm{a}}$:

$$\mathrm{p}K_{\mathrm{a}} = -\log(K_{\mathrm{a}})$$

Henderson–Hasselbalch equation:

$$\mathrm{pH} = \mathrm{p}K_{\mathrm{a}} + \log\frac{[\mathrm{base}]}{[\mathrm{acid}]}$$

10.2. Equilibrium and Thermodynamics

Equilibrium and free energy

$$K_{\text{eq}} = \exp(-\Delta G/RT)$$

Or for kcal/mol at room temperature

$$K_{\text{eq}} = 10^{(-0.75\Delta G)}$$

Relationship between standard free-energy change and the equilibrium constant:

$$\Delta G^\circ = -RT \ln K$$

Non-standard free energy for a reaction of form $a\text{A} + b\text{B} \longrightarrow c\text{C} + d\text{D}$, where $Q = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

11. Similarity of Equations

11.1. Linear Equations with Different Temperatures

Clausius–Clapeyron Equation:

$$\ln \frac{P_1}{P_2} = \frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{-\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Arrhenius Equation:

$$\ln \frac{k_1}{k_2} = \frac{E_{\text{a}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{-E_{\text{a}}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

van't Hoff Equation:

$$\ln \frac{K_1}{K_2} = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{-\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

van't Hoff Equation with K_{sp} and entropy, plot $\ln K_{\text{sp}}$ vs. $1/T$:

$$\Delta H^\circ - T\Delta S^\circ = \Delta G^\circ = -RT \ln K_{\text{sp}}$$

$$\ln K_{\text{sp}} = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

11.2. Summations

Standard entropy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta S^\circ_{\text{rxn}} = \sum nS^\circ(\text{products}) - \sum mS^\circ(\text{reactants})$$

Standard enthalpy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta H^\circ_{\text{rxn}} = \sum n\Delta H^\circ_{\text{f}}(\text{products}) - \sum m\Delta H^\circ_{\text{f}}(\text{reactants})$$

Standard free energy of reaction, where n and m are coefficients in the balanced reaction equation:

$$\Delta G^\circ_{\text{rxn}} = \sum n\Delta G^\circ_{\text{f}}(\text{products}) - \sum m\Delta G^\circ_{\text{f}}(\text{reactants})$$

12. Nuclear Decay and Mass Defect

Mass defect: add up the masses of protons and neutrons (for nuclei) and potentially electrons (for atoms), then subtract off the atomic mass.

$$\Delta m = [Z(m_{\mathrm{p}} + m_{\mathrm{e}}) + (A - Z)m_{\mathrm{n}}] - m_{\mathrm{atom}}$$

Table 6: Values for mass defect equation.

Variable	Description	Value
Δm	mass defect	amu
m_{p}	mass of a proton	1.007277
m_{n}	mass of a neutron	1.008665
m_{e}	mass of an electron	0.000548597
m_{atom}	mass of nuclide	${}^{A}_{Z}X$
Z	atomic number	number of protons
A	mass number	number of nucleons

Relationship between mass defect and energy released:

$$\Delta E = (\Delta m)c^2$$

Units:

- 6.022×10^{23} amu = 1 g
- 1 amu = 931.5 MeV
- J = kg·(m/s)^2
- 1 J = 6.242×10^{12} MeV

13. Compound Units

- joule = coulomb × volt
- joule = kg × m^2 / s^2 = mv^2
- joule = N × m = (kg × m / s^2) × m
- watt = joule / sec
- kilowatt = 1000 joules / sec
- kilowatt-hour = (1000 joules / sec) × 3600 sec = 3.6×10^6 joules
- amp = coulomb / second
- 1000 milliamp-hour = 1 amp-hour = 3600 amp-seconds = 3600 coulombs

14. Algebra

Quadratic Equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$ax^2 + bx + c = 0$$

15. Acidities

Acid	Conj. Base	pK_a	Remarks	
$HClO_4$	ClO_4^-	−11	perchloric acid	strong acids
HI	I^-	−10	hydroiodic acid	
HBr	Br^-	−9	hydrobromic acid	
H_2SO_4	HSO_4^-	−9	sulfuric acid	
HCl	Cl^-	−7	hydrochloric acid	
H_3O^+	H_2O	−1.7	hydronium	
HNO_3	NO_3^-	−1	nitric acid	
HSO_4^-	SO_4^{2-}	2	2nd H of sulfuric acid	weak acids
HF	F^-	3	hydrofluoric acid	
RCOOH	$RCOO^-$	4	carboxylic acids	
H_2S	HS^-	7	compare to H_2O	
NH_4^+	NH_3	9	ammonium	
$(RC(O))_2CH_2$	$(RC(O))_2CH^-$	9	α H of 1,3-diketones[a]	
HCN	NC^-	9.2	nitriles	
PhOH	PhO^-	10	phenols[b]	
$(ROC(O))_2CH_2$	$(ROC(O))_2CH^-$	13	α H of 1,3-diesters[a]	
H_2O	HO^-	15.7	water[c]	
ROH	RO^-	16	aliphatic alcohols[d]	
$RC(O)NH_2$	$RC(O)NH^-$	17	amides	
$HC(O)CH_3$	$HC(O)CH_2^-$	17	α H of aldehydes	
$RC(O)CH_3$	$RC(O)CH_2^-$	19.5	α H of ketones	
$ROC(O)CH_3$	$ROC(O)CH_2^-$	24	α H of esters	
$PhNH_2$	$PHNH^-$	25	aniline[e]	
$N{\equiv}C{-}CH_3$	$N{\equiv}C{-}CH_2^-$	25	α H of nitriles	
$R_2NC(O)CH_3$	$R_2NC(O)CH_2^-$	30	α H of amides	
$RC{\equiv}CH$	$RC{\equiv}C^-$	25	terminal alkyne (C_{sp}—H)	
H_2	H^-	36	hydrogen gas	
NH_3	NH_2^-	38	ammonia	
RNH_2	RNH^-	40	aliphatic amines	
$PhCH_3$	$PhCH_2^-$	41	benzylic (C_{sp^3} —H)	
C_6H_6	$C_6H_5^-$	42	aryl (C_{sp^2}—H)	
$H_2C{=}CHCH_3$	$H_2C{=}CHCH_2^-$	43	allylic (C_{sp^3}—H)	
$RCH{=}CH_2$	$RCH{=}CH^-$	47	vinylic (C_{sp^2}—H)	
RCH_3	RCH_2^-	50	alkanes (C_{sp^3}—H)	

[a] the α H of β-keto esters are ∼11

[b] depending on the substitution of Ph, pK_a can vary from 4–11

[c] water is amphoteric, so anything less acidic than water can act as a base in aqueous solution

[d] depending on the substitution of R, pK_a can vary from 15–17

[e] depending on the substitution of Ph, pK_a can vary from 18–28

A conjugate base can deprotonate any acid higher than it on this table.

Statistics Primer

John Terhorst

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1 Average

The arithmetic mean, or simply the mean, average, or expected value, is the sum of a collection of numbers divided by the count of numbers in the collection, Eq. 1.

$$\overline{x} = \frac{\sum_i (x_i)}{N} \tag{1}$$

1.1 Example

Let's imagine that a student, Manuel, is testing the accuracy of commercially available glucometers. Using a *PrecisionXtra* brand glucometer, he makes 15 measurements from the same glucose solution, Table 1. For this set of data, the average is calculated in Eq. 2.

Table 1: Manuel's *PrecisionXtra* data. All concentrations are in mg/dL.

Trial	[Glucose]
1	171
2	182
3	175
4	184
5	163
6	182
7	168
8	176
9	186
10	190
11	169
12	176
13	179
14	170
15	176

$$\overline{x} = \frac{\sum_i (x_i)}{15} = \frac{2647 \text{ mg/dL}}{15} = 176 \text{ mg/dL} \tag{2}$$

1.2 Usage

Excel and other spreadsheet programs provide an `AVERAGE()` function to automate this calculation.

2 Standard Deviation

The standard deviation is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range, Eq. 3.

$$\sigma = \sqrt{\frac{\sum_i (x_i - \overline{x})^2}{N-1}} \tag{3}$$

The standard deviation is calculated by:

1. Calculating the average (Section 1) of the set, $\overline{x}$. When calculating σ it is important not to round $\overline{x}$.
2. Calculating the difference between each point x_i and $\overline{x}$.
3. Squaring each difference.
4. Summing the squares.
5. Dividing by $N-1$, also known as the degrees of freedom (df), where N is the number of items in the set.
6. Taking the square root.

2.1 Example

Table 2 Shows how the steps above would be applied to Manuel's data from Table 1.

Table 2: Example calculation for standard deviation using data from Table 1, where $\overline{x} = 176.47$ mg/dL.

Trial	[Glucose]	$x_i - \overline{x}$	$(x_i - \overline{x})^2$	$\sum_i (x_i - \overline{x})^2$	σ
1	171	−5.47	29.88		
2	182	5.53	30.62		
3	175	−1.47	2.15		
4	184	7.53	56.75		
5	163	−13.47	181.35		
6	182	5.53	30.62		
7	168	−8.47	71.68		
8	176	−0.47	0.22		
9	186	9.53	90.88		
10	190	13.53	183.15		
11	169	−7.47	55.75		
12	176	−0.47	0.22		
13	179	2.53	6.42		
14	170	−6.47	41.82		
15	176	−0.47	0.22		
				781.73	7.47

2.2 Usage

Typically the *precision* of a repeated measurement is reported as $\overline{x} \pm \sigma$.

Given an expected Gaussian ("bell curve") distribution (see Figure 1 on page 16), the "average plus or minus one standard deviation" is the range within which 68.2% of the data

points are found. 95.4% of data points (an addition 27.2% of points) are found within the range $\overline{x} \pm 2\sigma$, and 99.7% of data points (an addition 4.3% of points) are found within the range $\overline{x} \pm 3\sigma$. Manuel could report the precision of the *PrecisionXtra* brand glucometer's measurements as 176 mg/dL $\pm$ 7.47 mg/dL to keep 3 significant figures.

Excel and other spreadsheet programs provide an `STDEV()` function to automate this calculation.

3 Confidence Interval

Now that we have evaluated the precision of the data, we should evaluate how accurate they are: how well they agree with the actual concentration of glucose in the solution.

The confidence interval (CI) is a type of estimate that proposes a range of plausible values for an unknown parameter—for example, the mean—from a sample. This range, μ, is known as a *true mean* and depends upon a parameter t that is itself a function of the size of the sample and the desired level of confidence in μ. The formula is given in Eq. 4.

$$\mu = \overline{x} \pm \frac{t\sigma}{\sqrt{N}} \tag{4}$$

Tables of t are available online, e.g., https://tinyurl.com/statsttable. Most commonly a 95% confidence level is used, but 90% and 99% are also common.

3.1 Example

Let's apply this to Manuel's data set. We already know $\overline{x}$, σ, and N. Looking up t on the linked table, we find that for a sample size $N = 15$ (df $= N - 1 = 14$), $t = 2.145$ at the 95% confidence level. Thus:

$$\mu = 176 \pm \frac{(2.145)(7.47)}{\sqrt{15}} = 176 \text{ mg/dL} \pm 4.14 \text{ mg/dL} \tag{5}$$

3.2 Usage

There are two ways to interpret this range:

1. You can be 95% confident that the *actual value* of the concentration of glucose in the solution is between 171.9 mg/dL and 180.1 mg/dL.
2. If the actual value of the concentration of glucose in the solution is known to be within the range of 171.9 mg/dL and 180.1 mg/dL, then you can say with 95% confidence that your testing method is accurate.

4 Outliers

The concept of a normal Gaussian distribution, assumed in these methods, raises the question of how to handle outliers; that is, what to do with individual values that are much farther away from the mean than any of the other values. Outliers can be the result of experimental error, sampling problems, or other random events. Keeping outliers in a data set can obscure

reliability of and confidence in the data, but how do we know when we can discard a data point? There are two common ways to decide if a data point can be ignored: the Grubbs' test and Dixon's Q-test.

Any outlier will always be the highest or lowest value in a data set, so before we begin, let's sort Manuel's data numerically, Table 3.

Table 3: Extreme endpoints of Manuel's *PrecisionXtra* data. All concentrations are in mg/dL.

Trial	[Glucose]
1	163
2	168
:	:
14	186
15	190

4.1 Grubbs' Test

Like the CI test for true mean, Grubbs' test (Eqs. 6 and 7) also relies on a parameter Z_{ref} that is based on sample size and confidence level. A table for Z_{ref} can be found at https://tinyurl.com/GrubbsZvalues.

$$Z = \frac{|x_i - \overline{x}|}{\sigma} \tag{6}$$

$$\text{If } Z > Z_{\text{ref}} \text{ then } x_i \text{ can be considered an outlier} \tag{7}$$

Let's test the highest measurement (190 mg/dL, Eq. 8) and the lowest measurement (163 mg/dL, Eq. 9) made by Manuel. For the data set, $N = 15$, so at the 95% confidence level (the 0.05 column) we find $Z_{\text{ref}} = 2.37$. As before, $\overline{x} = 176$ mg/dL and $\sigma = 7.47$ mg/dL.

$$Z = \frac{|190 - 176|}{7.47} = 1.87 \tag{8}$$

$$Z = \frac{|163 - 176|}{7.47} = 1.74 \tag{9}$$

Since neither $Z > Z_{\text{ref}}$ (Eq. 7), neither of these data points should be ignored.

4.2 Dixon's Q-Test

Like the CI test for true mean and Grubbs' test, Dixon's Q-test (Eqs. 10 and 11) also relies on a parameter Q_{ref} that is based on sample size and confidence level. A table for Q_{ref} can be found at https://tinyurl.com/DixonQvalues.

$$Q = \frac{\text{gap}}{\text{range}} = \frac{|\text{suspected outlier} - \text{closest value}|}{(\text{maximum value} - \text{minimum value})} \tag{10}$$

$$\text{If } Q > Q_{\text{ref}} \text{ then } x_i \text{ can be considered an outlier} \tag{11}$$

Let's again test the highest measurement (190 mg/dL, Eq. 12) and the lowest measurement (163 mg/dL, Eq. 13) made by Manuel. For the data set, $N = 15$, so at the 95% confidence level we find $Q_{\text{ref}} = 0.34$.

$$Q = \frac{|190 - 186|}{(190 - 163)} = 0.148 \tag{12}$$

$$Q = \frac{|163 - 168|}{(190 - 163)} = 0.185 \tag{13}$$

Since neither $Q > Q_{\text{ref}}$ (Eq. 11), neither of these data points should be ignored.

4.3 What would an outlier look like in a data set?

Let's consider another set of measurements made by another student, Jake, using the same glucose solution and the same model *PrecisionXtra* brand glucometer as Manuel but not the exact same instrument. Data are given in Table 4, sorted from lowest measurement to highest, and analysis is given in Table 5. As is clear, the 140 mg/dL data point is an outlier: $Z_{\text{low}} > Z_{\text{ref}}$ and $Q_{\text{low}} > Q_{\text{ref}}$.

Table 4: Jake's *PrecisionXtra* data. All concentrations are in mg/dL.

Trial	[Glucose]
1	140
2	168
3	170
4	172
5	172
6	173
7	174
8	175
9	176
10	176
11	177
12	177
13	185
14	188
15	191
$\overline{x}$	174
σ	11.49

Table 5: Statistical parameters for Jake's *PrecisionXtra* data, based on Table 4. Z_{low} is the Grubbs' test on the lowest data point in the set, Z_{high} is the Grubbs' test on the highest data point in the set, Q_{low} is the Q-test on the lowest data point in the set, and Q_{high} is the Q-test on the highest data point in the set.

Stat	Value
Z_{low}	2.98
Z_{high}	1.46
Z_{ref}	2.37
Q_{low}	0.55
Q_{high}	0.06
Q_{ref}	0.34

5 *t*-Test, one- and two-tailed

We now turn to statistical tests that allow us to compare two sets of data for likeness. This type of analysis is important in science because it allows us to compare the significance of a variable against a control group. These tests test a *null hypothesis*, which is typically that the means two sets of data are similar; in other words, that the variable being tested has *no effect* on the means of the data. At their most fundamental level, these tests ask, "Did the variable change the mean?" And the response is a number that tells us how confident we can be that the answer is, "No."

The most common of this type of test is the t-test, which comes in several flavors, including "one-tailed" tests and "two-tailed" tests. By "tails" we mean the extreme ends of the distribution curve[1] (the highest and lowest 2.5% of data points when working at the 95% confidence level, to be clear). When using a two-tailed t-test, you are testing for similarity given the possibility of the relationship deviating in either direction: for example, if we are comparing a student's exam score (x) against the average exam score of the class ($\overline{x}$), a two-tailed test would allow for the possibility that $x \ll \overline{x}$ or that $x \gg \overline{x}$. A one-tailed t-test, on the other hand, assumes that the deviation of x from $\overline{x}$ can only happen in one tail of the distribution curve (the highest or lowest 5% of data points, *but not both*). The one-tailed test provides more power to detect an effect in one direction by not testing the effect in the other direction.

Imagine you are testing a new drug that you believe is an improvement over an existing drug. You wish to maximize your ability to detect the improvement, so you opt for a one-tailed test. In doing so, you fail to test for the possibility that the new drug is *less* effective than the existing drug; that effect could be missed, and so a one-tailed test would not be appropriate. Say, on the other hand, that you have developed a new drug that is cheaper than the existing drug. In testing this drug, you do not care if it is significantly more effective; you only wish to show that it is not less effective. In this scenario, a one-tailed test would be appropriate.

5.1 *t*-Test assuming equal variances

Let's compare Manuel's data (Table 1) to Jake's data (Table 4, with the outlier removed). The question we would like to answer is, "Are the two glucometers equally reliable at

[1] Usually a Gaussian, see Figure 1 on page 16.

measuring blood glucose levels?" Our null hypothesis is, "There is no difference between the glucometers." In this test, we can assume *equal variances*[2] because the skill of the user should not affect measurement results. We should use a two-tailed test, because deviation in either direction would mean that we reject the null hypothesis. Let's look at the data side by side, Table 6.

Table 6: Two sets of *PrecisionXtra* data. All concentrations are in mg/dL.

	[Glucose]	
Trial	Manuel	Jake
1	163	168
2	168	170
3	169	172
4	170	172
5	171	173
6	175	174
7	176	175
8	176	176
9	176	176
10	179	177
11	182	177
12	182	185
13	184	188
14	186	191
15	190	—

The output of the t-test from Excel is given in Table 7. Details on how to run the test are given in Section 5.1.1. Note that the variances are similar and that both one-tailed and two-tailed analyses are performed simultaneously.

To reiterate, the null hypothesis for this test is that there is no difference between the two sets of data (in an experiment we might say that the null hypothesis is that there is no difference between the control group data and the experimental group data). The decisive value $P(T \leq t) = 0.926$ in Table 7 indicates a 92.6% chance that the null hypothesis is correct (that is, that we *accept* the null hypothesis), or, in other words, a 7.4% chance that that there is a statistically significant difference between the two sets of data (that is, that we reject the null hypothesis). In practice, a 95% confidence level is required to reject the null hypothesis—so we would need to get a $P \leq 0.05$ to reject the null hypothesis. Yes, this is a very high bar! Data must be extremely conclusive before the null hypothesis can be rejected. Thus, our data do not persuade us to conclude that Manuel's and Jake's glucometers are significantly different.

5.1.1 Usage

In Excel, choose Data → Data Analysis → t-Test: Two Sample Assuming Equal Variances. Select the control group data for Variable 1 and the experimental group data for Variable

[2] A more technical way to frame this is that this test assumes that both groups of data are sampled from Gaussian populations with the *same standard deviation*.

Table 7: Excel output for two-sample t-test assuming equal variances.

	Variable 1	*Variable 2*
Mean	176.47	176.71
Variance	55.84	45.45
Observations	15	14
Pooled Variance	50.84	
Hypothesized Mean Difference	0	
df	27	
t Stat	-0.093	
$P(T \leq t)$ one-tail	0.463	
t Critical one-tail	1.70	
$P(T \leq t)$ two-tail	0.926	$\longleftarrow$
t Critical two-tail	2.05	

2, enter 0 for the Hypothesized Mean Difference, and keep 0.05 for Alpha. Then click OK. Your output should look similar to Table 7.

5.2 *t*-Test assuming unequal variances

As mentioned in Section 5.1, we used the t-test assuming equal variances because we could assume that both glucometers had the same standard deviation. There is another flavor of t-test that assumes *unequal* variances between data sets, yet the null hypothesis remains the same—that there is no meaningful difference in the means. This is also called a Welch's t-test. Interestingly, Sawilowsky[3] points out that this situation simply doesn't often come up in science, because it suggests that more than one variable is at play—not a good setup for an experiment. To illustrate where this might be useful, we could compare dividend yields for the top ten NYSE and NASDAQ stocks. Since each data set would contain a different set of ten stocks, it would not be possible to assume equal variances.

In the context of our glucometer experiment, we might compare data taken by Manuel with a *PrecisionXtra* glucometer and by Jake with an *AccuCheck* glucometer. Since the glucometers are of different brands and the sample was measured by different people and the data sets have a different number of entries, the data sets they produce are likely not to have the same standard deviation. Let's look, Table 8.

The output of the t-test from Excel is given in Table 9. Details on how to run the test are given in Section 5.2.1. Note that the variances are *different* and that both one-tailed and two-tailed analyses are performed simultaneously.

To reiterate, the null hypothesis for this test is that there is no difference between the two sets of data (in an experiment we might say that the null hypothesis is that there is no difference between the control group data and the experimental group data). The decisive value $P(T \leq t) = 0.904$ in Table 9 indicates a 90.4% chance that the null hypothesis is correct (that is, that we *accept* the null hypothesis), or, in other words, a 9.6% chance that that there is a statistically significant difference between the two sets of data (that is, that we

[3]Sawilowsky, S. S. "Fermat, Schubert, Einstein, and Behrens–Fisher: The Probable Difference Between Two Means With Different Variances." *J. Modern Appl. Stat. Meth.* **2002**, *1*, 461–472.

Table 8: Data to be used in a t-Test assuming unequal variances.

Manuel *PrecisionXtra*	Jake *AccuCheck*
163	160
168	160
169	161
170	163
171	165
175	167
176	168
176	168
176	168
179	170
182	170
182	173
184	175
186	178
190	179
	180
	184
	188
	189
	189
	190
	190
	193
	195
	198

Table 9: Excel output for two-sample t-test assuming unequal variances.

	Variable 1	*Variable 2*
Mean	176.47	176.84
Variance	55.83	144.39
Observations	15	25
Hypothesized Mean Difference	0	
df	38	
t Stat	-0.121	
$P(T \leq t)$ one-tail	0.452	
t Critical one-tail	1.68	
$P(T \leq t)$ two-tail	0.904	$\longleftarrow$
t Critical two-tail	2.02	

reject the null hypothesis). In practice, a 95% confidence level is required to reject the null hypothesis—so we would need to get a $P \leq 0.05$ to reject the null hypothesis. Yes, this is a very high bar! Data must be extremely conclusive before the null hypothesis can be rejected. Thus, our data do not persuade us to conclude that Manuel's and Jake's glucometers are significantly different.

5.2.1 Usage

In Excel, choose Data → Data Analysis → t-Test: Two Sample Assuming Unequal Variances. Select the control group data for Variable 1 and the experimental group data for Variable 2, enter 0 for the Hypothesized Mean Difference, and keep 0.05 for Alpha. Then click OK. Your output should look similar to Table 9.

5.3 Paired t-Test

The third variation of the t-test is the paired t-test. This test is for paired, correlated, or dependent samples and compares means of matched pairs of scores on some variable. A common example might be if you measured performance on some task prior to training, and the same group of people were again measured for task performance after training. (You do have to make some assumptions about the nature of the scores, however. Can you think of an example?) Note that this test *cannot* assume equal variance, because the result of the training may make the scores more clustered and therefore lower the standard deviation.

Let's look at an example where a group of employees is tested on a certain topic both before and after training, Table 10. Did the training affect the test results? At a cursory glance, we'd say yes. But was the improvement significant, or could it be a fluke? The analysis is given in Table 11. Here, let's first notice that the variance decreased significantly because scores got more clustered. Let's also consider whether a one-tailed or two-tailed test is best. Do we care *only* if training improved scores? Or do we also care if training might have worsened scores? We probably don't care if training worsened scores, but we can also see that all the scores went up—so a two-tailed test doesn't make sense here. Let's put all the power of the test into *only* determining whether the *improvement* of the scores was significant.

To reiterate, the null hypothesis for this test is that there is no difference between the two sets of data (in an experiment we might say that the null hypothesis is that there is no difference between the control group data and the experimental group data). The decisive value $P(T \leq t) = 0.007$ in Table 11 indicates a 0.7% chance that the null hypothesis is correct (that is, that we *accept* the null hypothesis), or, in other words, a 99.3% chance that that there is a statistically significant difference between the two sets of data (that is, that we reject the null hypothesis). In practice, a 95% confidence level is required to reject the null hypothesis—so we need to get a $P \leq 0.05$ to reject the null hypothesis. Thus here we can reject the null hypothesis and conclude that the training was meaningful.

While we're on the subject, let's look at $P(T \leq t)$ one-tail (0.007) compared to $P(T \leq t)$ two-tail (0.015) in Table 11. Both tests let us reject the null hypothesis, but by ignoring the possibility that training might have worsened scores and only analyzing whether training significantly improved scores, we can be much more confident in our conclusion to reject the null hypothesis.

Table 10: Exam results before and after training.

Employee	Before	After
Anna	68	75
Bob	75	79
Carl	85	88
Dave	71	72
Erin	92	99
Fred	50	80
Gale	42	75
Henry	77	88
Ian	89	95
Joanne	55	60
Karla	99	100

Table 11: Excel output for the paired two-sample t-test.

	Variable 1	*Variable 2*
Mean	73	82.82
Variance	330	154.2
Observations	11	11
Pearson Correlation	0.799	
Hypothesized Mean Difference	0	
df	10	
t Stat	-2.93	
$P(T \leq t)$ one-tail	0.007	$\longleftarrow$
t Critical one-tail	1.81	
$P(T \leq t)$ two-tail	0.015	
t Critical two-tail	2.23	

5.3.1 Usage

In Excel, choose Data → Data Analysis → t-Test: Paired Two Sample for Means. Select the control group data for Variable 1 and the experimental group data for Variable 2, enter 0 for the Hypothesized Mean Difference, and keep 0.05 for Alpha. Then click OK. Your output should look similar to Table 11.

6 F-Test

Whereas the t-test (Sections 5.1–5.3) tests for similarity of means, the F-test is a test for the null hypothesis that two normal populations have the same *variance*. For example, if we are testing a new process, we may be interested in knowing if the new process is more or less variable than the old process. The F-test comes in one- and two-tailed varieties and the selection for one or the other works the same as in the t-test. If we are testing a new process for data collection, a two-tailed test is appropriate if we care if the new process is more *or* less variable than the old one. If we have a new process for data collection that is faster than the old process, we might only care that the new process is not more variable, so a one-tailed test would be better. Unfortunately, in Excel only the one-tailed variety is available, so we'll go through how to use it properly.

6.1 Example

In Sections 5.1 and 5.2 we assumed equal and unequal variances, respectively. Using an F-test, we could have first checked first which version was warranted! Let's test for variance in Table 6, where we assumed equal variance. Output is given in Table 12. Usage is given in Section 6.2.

Table 12: F-test output based on data in Table 6.

	Variable 1	*Variable 2*
Mean	176.46	176.71
Variance	55.84	45.45
Observations	15	14
df	14	13
F	1.23	
$P(F \leq f)$ one-tail	0.358	⟵
F Critical one-tail	2.55	

As with our t-test, the F-test tests the null hypothesis. The decisive value $P(F \leq f) = 0.358$ indicates a 35.8% confidence in our null hypothesis—but remember, P must be ≤ 0.05 to reject the null hypothesis, so here we cannot reject the null hypothesis. We conclude the samples do not have significantly different variances. Now we know that we used the correct t-test in Section 5.1!

If we repeat the F-test using the data in Table 8 where we assumed unequal variance, we find $P(F \leq f) = 0.034$, which indicates a 3.4% confidence in our null hypothesis. Since $P \leq 0.05$, we *reject* the null hypothesis and conclude that the variances of the two data sets are unequal. Now we know that we used the correct t-test in Section 5.2!

6.2 Usage

In Excel, choose Data → Data Analysis → F-Test Two-Sample for Variances. Select the control group data for Variable 1 and the experimental group data for Variable 2, and keep 0.05 for Alpha. Then click OK. Your output should look similar to Table 12. **Note: it is necessary that the variance for Variable 1 be greater than the variance for Variable 2, so if you run your F-test and end up with a greater variance for Variable 2, you need to swap your data and re-run the test.**

7 ANOVA

The ANOVA (ANalysis Of VAriance) test is used to analyze the means and variances of groups within a sample. Somewhat combining our t-test and F-test, ANOVA tests two null hypotheses: that there is no difference of both *means* and *variances* between groups. It is different than the t- and F-test in that it can be used to compare more than two data sets.

7.1 Example

Let's say we do an enzyme activity assay on four different parts of the brain (cortex, cerebellum, brainstem, and basal forebrain). Is the enzyme more active in any part of the brain? As you see, a t- or F-test doesn't really make sense here. Instead, we can use an ANOVA test. Let's look at some sample data, Table 13.

Table 13: Enzyme reaction rates (ΔA/min) for trials within each brain region.

Trial	Cortex	Cerebellum	Brainstem	Basal Forebrain
1	0.123	0.032	0.061	0.239
2	0.115	0.040	0.051	0.250
3	0.116	0.030	0.098	0.215

In this data set, we can ask (a) how reliable is the data amongst the three trials for each tissue type? and (b) how does the data compare between the different tissue types? Both of these questions (and more) can be answered by an ANOVA analysis. Output is given in Tables 14 and 15 and usage is given in Section 7.2.

In Table 14, the first-tier output is broken into two sections: trials (rows from Table 13) and brain regions (columns from Table 13). In this particular case, the per-trial (row) analysis is not useful or interesting (e.g., we don't care how Trial 1 varied from region to region) *but that might not be true for different types of data*—so you can see how ANOVA can be a very powerful tool. Instead, what we care about is the per-region (column) analysis and how different regions (columns) compare.

In Table 15 is given the second-tier output for trials (rows) and brain regions (columns). Again in this case we can ignore the trials analysis and focus on the brain regions. Recall that the ANOVA tests the null hypotheses that there is no difference of both *means* and *variances* between regions. The decisive values are $P = 4.28 \times 10^{-5}$, $F = 93.4879$ and $F_{\text{crit}} = 4.0661$. $P = 4.28 \times 10^{-5} \ll 0.05$ indicates that we must reject the null hypothesis of means—our data is *significant* and the means between brain regions are indeed different. Furthermore, since $F > F_{\text{crit}}$, we conclude that we *cannot* reject the null hypothesis of *variances*, which in this context signifies good data.

Table 14: ANOVA analysis of Table 13.

Category	Count	Sum	Average	Variance
Trial 1	4	0.456	0.114	8.41×10^{-3}
Trial 2	4	0.457	0.114	9.31×10^{-3}
Trial 3	4	0.459	0.115	5.84×10^{-3}
Cortex	3	0.354	0.118	1.90×10^{-5}
Cerebellum	3	0.102	0.034	2.80×10^{-5}
Brainstem	3	0.210	0.070	6.13×10^{-4}
Basal Forebrain	3	0.705	0.235	3.20×10^{-4}

Table 15: ANOVA analysis of Table 13 based on Table 14.

Source	SS	df	F	P-value	F_{crit}
Trials	2.17×10^{-6}	2	1.38×10^{-4}	0.9998	4.2565
Regions	0.06873	3	93.4879	4.28×10^{-5}	4.0661
Total	0.07069	11			

7.2 Usage

In Excel, organize your data per Table 13. Choose Data → Data Analysis → Anova Single Factor. For Input Range, select your data including column and row headings, and make sure to tick the Labels box. For analysis of rows, choose Grouped by: Rows. For analysis of columns, choose Grouped by: Columns. If your analysis would benefit from both types of analysis (for example, if Manuel, Jake, and Heather each performed one trial, you may care to see if data varied significantly from person to person), then run one type and repeat with the other type.

8 Appendix

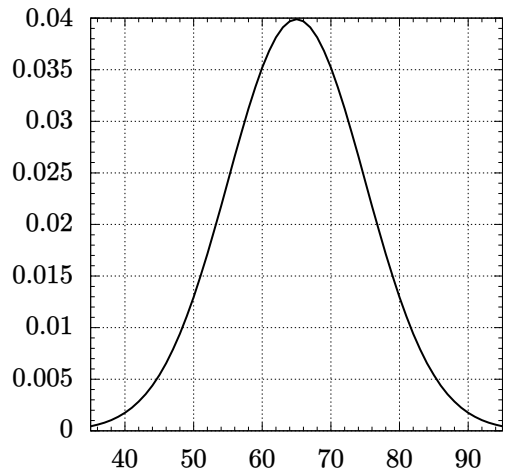

Figure 1: A Gaussian (bell curve) distribution with $\overline{x} = 65$ and $\sigma = 10$.

Standand Enthalpies of Formation & Standard Entropies of Common Compounds

Substance	*State*	ΔH_f° $(\frac{\text{kJ}}{\text{mol}})$	S° $(\frac{\text{J}}{\text{mol}\cdot\text{K}})$
Ag	s	0	42.6
Ag^+	aq	105.79	72.7
AgCl	s	−127.01	96.2
AgBr	s	−100.4	107.1
$AgNO_3$	s	−124.4	140.9
Al	s	0	28.3
Al^{+3}	aq	−538.4	−321.7
$AlCl_3$	s	−704	110.7
Al_2O_3	s	−1675.7	50.9
Ba	s	0	62.8
$BaCl_2$	s	−858.6	123.7
$BaCO_3$	s	−1216.3	112.1
$Ba(NO_3)_2$	s	−992	214
BaO	s	−553.5	70.4
$Ba(OH)_2$	s	−998.2	112
$BaSO_4$	s	−1473.2	132.2
Br_2	ℓ	0	152.2
C	s	0	5.7
CCl_4	ℓ	−135.4	216.4
$CHCl_3$	ℓ	−134.5	201.7
CH_4	g	−74.8	186.2
C_2H_2	g	+226.7	200.8
C_2H_4	g	+52.3	219.5
C_2H_6	g	−84.7	229.5
C_3H_8	g	−103.8	269.9
CH_3OH	ℓ	−238.7	126.8
C_2H_5OH	ℓ	−277.7	160.7
CO	g	−110.53	197.6
CO_2	g	−393.51	213.6
${CO_3}^{-2}$	aq	−675.23	−56.9
Ca	s	0	41.4
Ca^{+2}	aq	−543.0	−53.1
$CaCl_2$	s	−795.8	104.6
$CaCO_3$	s	−1206.9	92.9
CaO	s	−634.92	39.8
$Ca(OH)_2$	s	−986.1	83.4
$CaSO_4$	s	−1434.1	106.7
Cd	s	0	51.8
Cd^{+2}	aq	−75.92	−73.2
$CdCl_2$	s	−391.5	115.3
CdO	s	−258.35	54.8

Substance	*State*	ΔH_f° $(\frac{\text{kJ}}{\text{mol}})$	S° $(\frac{\text{J}}{\text{mol}\cdot\text{K}})$
Cl_2	g	0	223.0
Cl^-	aq	−167.080	56.5
${ClO_4}^-$	aq	−128.10	182.0
Cr	s	0	23.8
Cr_2O_3	g	−1139.7	81.2
Cu	s	0	33.2
Cu^+	aq	+71.7	40.6
Cu^{+2}	aq	+64.8	−99.6
CuO	s	−157.3	42.6
Cu_2O	s	−168.6	93.1
CuS	s	−53.1	66.5
Cu_2S	s	−79.5	120.9
$CuSO_4$	s	−771.4	107.6
F^-	aq	−335.35	−13.8
F_2	g	0	202.7
Fe	s	0	27.3
$Fe(OH)_3$	s	−823.0	106.7
Fe_2O_3	s	−824.2	87.4
Fe_3O_4	s	−1118.4	146.4
H_2	g	0	130.6
H^+	aq	0	0.0*
HBr	g	−36.29	198.6
${HCO_3}^-$	aq	−689.93	91.2
HCl	g	−92.31	186.8
HF	g	−273.30	173.7
HI	g	26.50	206.5
HNO_3	ℓ	−174.1	155.6
${HPO_4}^{-2}$	aq	−1299.0	−33.5
${HSO_4}^-$	aq	−886.9	131.8
H_2O	ℓ	−285.830	69.9
H_2O	g	−241.826	188.7
${H_2PO_4}^-$	aq	−1302.6	90.4
H_2S	g	−20.6	205.7
Hg	ℓ	0	76.0
Hg^{+2}	aq	170.21	−32.2
HgO	cr,red	−90.79	70.3

*The standard entropy of the $H^+(aq)$ ion is defined to be 0.

Substance	*State*	ΔH_f° $(\frac{\text{kJ}}{\text{mol}})$	S° $(\frac{\text{J}}{\text{mol}\cdot\text{K}})$
I^-	aq	−56.78	111.3
I_2	s	0	116.1
K	s	0	64.2
K^+	aq	−252.14	102.5
KBr	s	−393.8	95.9
KCl	s	−436.7	82.6
$KClO_3$	s	−397.7	143.1
$KClO_4$	s	−432.8	151.0
KNO_3	s	−494.6	133.0
Mg	s	0	32.7
Mg^{+2}	aq	−467.0	−138.1
$MgCl_2$	s	−641.3	89.6
$MgCO_3$	s	−1095.8	65.7
MgO	s	−601.60	26.9
$Mg(OH)_2$	s	−924.5	63.2
$MgSO_4$	s	−1284.9	91.6
Mn	s	0	32.0
Mn^{+2}	aq	−220.8	−73.6
MnO	s	−385.2	59.7
MnO_2	s	−520.0	53.0
N_2	g	0	191.5
NH_3	g	−45.94	192.3
$NH_4{}^+$	aq	−133.26	113.4
$NO_2{}^-$	aq	−104.6	123.0
$NO_3{}^-$	aq	−206.85	146.4
N_2H_4	ℓ	+50.6	121.2
NH_4Cl	s	−314.4	94.6
NH_4NO_3	s	−365.6	151.1
NO	g	+90.2	210.7
NO_2	g	+33.2	240.0
N_2O_4	g	+9.2	304.2
Na	s	0	51.2
Na^+	aq	−240.34	59.0
NaCl	s	−411.2	72.1
NaF	s	−573.6	51.5
NaOH	s	425.6	64.5

Substance	*State*	ΔH_f° $(\frac{\text{kJ}}{\text{mol}})$	S° $\frac{\text{J}}{\text{mol K}}$
Ni	s	0	29.9
NiO	s	−239.7	38.0
OH^-	aq	−230.015	−10.8
O_2	g	0	205.0
P_4	s	0	164.4
PCl_3	g	−287.0	311.7
PCl_5	g	−374.9	364.5
$PO_4{}^{-3}$	aq	−1277.4	−222
Pb	s	0	64.8
Pb^{+2}	aq	0.92	10.5
$PbBr_2$	s	−278.7	161.5
$PbCl_2$	s	−359.4	136.0
PbO	s	−219.0	66.5
PbO_2	s	−277.4	68.6
S	s	0	31.8
SO_2	g	−296.81	248.1
SO_3	g	−395.7	256.7
$SO_4{}^{-2}$	aq	−909.34	20.1
$S_2{}^-$	aq	+33.1	−14.6
Si	s	0	18.8
SiO_2	s	−910.7	41.8
Sn	s	0	51.6
Sn^{+2}	aq	−8.9	−17.4
SnO_2	s	−577.63	52.3
Zn	s	0	41.6
Zn^{+2}	aq	−153.39	−112.1
ZnI_2	s	−208.0	161.1
ZnO	s	−350.46	43.6
ZnS	s	−206.0	57.7

TABLE 9.3 Average Bond Energies

Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
H—H	436	N—N	163	Br—F	237
H—C	414	N═N	418	Br—Cl	218
H—N	389	N≡N	946	Br—Br	193
H—O	464	N—O	222	I—Cl	208
H—S	368	N═O	590	I—Br	175
H—F	565	N—F	272	I—I	151
H—Cl	431	N—Cl	200	Si—H	323
H—Br	364	N—Br	243	Si—Si	226
H—I	297	N—I	159	Si—C	301
C—C	347	O—O	142	S—O	265
C═C	611	O═O	498	Si═O	368
C≡C	837	O—F	190	S═O	523
C—N	305	O—Cl	203	Si—Cl	464
C═N	615	O—I	234	S═S	418
C≡N	891	F—F	159	S—F	327
C—O	360	Cl—F	253	S—Cl	253
C═O	736*	Cl—Cl	243	S—Br	218
C≡O	1072			S—S	266
C—Cl	339				

*799 in CO_2

TABLE 10.4 Van der Waals Constants for Common Gases

Gas	a ($L^2 \cdot atm/mol^2$)	b (L/mol)
He	0.0342	0.02370
Ne	0.211	0.0171
Ar	1.35	0.0322
Kr	2.32	0.0398
Xe	4.19	0.0511
H_2	0.244	0.0266
N_2	1.39	0.0391
O_2	1.36	0.0318
Cl_2	6.49	0.0562
H_2O	5.46	0.0305
CH_4	2.25	0.0428
CO_2	3.59	0.0427
CCl_4	20.4	0.1383

Values of Grubbs Statistic (G)

Number of Observations n	Confidence Level (%)					
	99.9	99.5	99	97.5	95	90
3	1.155	1.155	1.155	1.155	1.153	1.148
4	1.499	1.496	1.492	1.481	1.463	1.425
5	1.780	1.764	1.749	1.715	1.672	1.602
6	2.011	1.973	1.944	1.887	1.822	1.729
7	2.201	2.139	2.097	2.020	1.938	1.828
8	2.358	2.274	2.221	2.126	2.032	1.909
9	2.492	2.387	2.323	2.215	2.110	1.977
10	2.606	2.482	2.410	2.290	2.176	2.036
11	2.705	2.564	2.485	2.355	2.234	2.088
12	2.791	2.636	2.550	2.412	2.285	2.134
13	2.867	2.699	2.607	2.462	2.331	2.175
14	2.935	2.755	2.659	2.507	2.371	2.213
15	2.997	2.806	2.705	2.549	2.409	2.247
16	3.052	2.852	2.747	2.585	2.443	2.279
17	3.103	2.894	2.785	2.620	2.475	2.309
18	3.149	2.932	2.821	2.651	2.504	2.335
19	3.191	2.968	2.854	2.681	2.532	2.361
20	3.230	3.001	2.884	2.709	2.557	2.385
30	3.507	3.236	3.103	2.908	2.745	2.563
40	3.673	3.381	3.240	3.036	2.866	2.682
50	3.789	3.483	3.336	3.128	2.956	2.768
60	3.874	3.560	3.411	3.199	3.025	2.837
70	3.942	3.622	3.471	3.257	3.082	2.893
80	3.998	3.673	3.521	3.305	3.130	2.940
90	4.044	3.716	3.563	3.347	3.171	2.981
100	4.084	3.754	3.600	3.383	3.207	3.017

Source: ASTM E178-00, “Standard Practice for Dealing with Outlying Observations”

Values of Student's t

Degrees of Freedom	Confidence Level (%)						
	50	90	95	97.5	99	99.5	99.9
1	1.000	6.314	12.706	25.452	63.657	127.321	636.619
2	0.816	2.920	4.303	6.205	9.925	14.089	31.599
3	0.765	2.353	3.182	4.177	5.841	7.453	12.924
4	0.741	2.132	2.776	3.495	4.604	5.598	8.610
5	0.727	2.015	2.571	3.163	4.032	4.773	6.869
6	0.718	1.943	2.447	2.969	3.707	4.317	5.959
7	0.711	1.895	2.365	2.841	3.499	4.029	5.408
8	0.706	1.860	2.306	2.752	3.355	3.833	5.041
9	0.703	1.833	2.262	2.685	3.250	3.690	4.781
10	0.700	1.812	2.228	2.634	3.169	3.581	4.587
11	0.697	1.796	2.201	2.593	3.106	3.497	4.437
12	0.695	1.782	2.179	2.560	3.055	3.428	4.318
13	0.694	1.771	2.160	2.533	3.012	3.372	4.221
14	0.692	1.761	2.145	2.510	2.977	3.326	4.140
15	0.691	1.753	2.131	2.490	2.947	3.286	4.073
16	0.690	1.746	2.120	2.473	2.921	3.252	4.015
17	0.689	1.740	2.110	2.458	2.898	3.222	3.965
18	0.688	1.734	2.101	2.445	2.878	3.197	3.922
19	0.688	1.729	2.093	2.433	2.861	3.174	3.883
20	0.687	1.725	2.086	2.423	2.845	3.153	3.850
30	0.683	1.697	2.042	2.360	2.750	3.030	3.646
40	0.681	1.684	2.021	2.329	2.704	2.971	3.551
50	0.679	1.676	2.009	2.311	2.678	2.937	3.496
60	0.679	1.671	2.000	2.299	2.660	2.915	3.460
70	0.678	1.667	1.994	2.291	2.648	2.899	3.435
80	0.678	1.664	1.990	2.284	2.639	2.887	3.416
90	0.677	1.662	1.987	2.280	2.632	2.878	3.402
100	0.677	1.660	1.984	2.276	2.626	2.871	3.390
∞	0.674	1.645	1.960	2.241	2.576	2.807	3.291

Values for the F Statistic at the 95% Confidence Level

Degrees of Freedom (denominator)	Degrees of Freedom (numerator)										
	2	3	4	5	6	7	8	9	10	20	∞
2	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.45	19.50
3	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.66	8.53
4	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.80	5.63
5	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.56	4.36
6	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	3.87	3.67
7	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.44	3.23
8	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.15	2.93
9	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	2.94	2.71
10	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.77	2.54
20	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.12	1.84
∞	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.57	1.00

Values for the F Statistic at the 90% Confidence Level

Degrees of Freedom (denominator)	Degrees of Freedom (numerator)										
	2	3	4	5	6	7	8	9	10	20	∞
2	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.44	9.49
3	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.18	5.13
4	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.84	3.76
5	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.21	3.10
6	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.84	2.72
7	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.59	2.47
8	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.42	2.29
9	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.30	2.16
10	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.20	2.06
20	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.79	1.61
∞	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.42	1.00